

Notes on the code port

1 How the code was ported

The code was ported manually but given today's editors the bulk of the work could be done with a relatively small set of scripts.

In a few cases (e.g. about1, about2) I also had to modify .RC files. Typically these modifications are also acceptable for the original C/C++ compilers. Most of the time the fpc-supplied wrd does the job; for the hexcalc programs however I used IBM RC.EXE.

It was quite helpful to have source code available for several units (fpc, Virtual Pascal; and sometimes other compilers as well).

2 Priorities

I wanted to get Pascal code up and running without paying attention to “all” details. Along the road several decisions had to be made, and a lot of them are somewhat arbitrary. I did try to be consistent though.

By and large I did not try to “improve” the programs.

An example of a decision: C is case sensitive, pascal is not. So what to do with e.g.: HPS hps? Two options (and I used both):

- introduce an underscore in the variable name, e.g.: h_ps: HPS;
- introduce a Pascal type for HPS, e.g.: hps: cardinal;

3 Compiling with C

Sometimes it was just handy to play with the original C code, even more so because some of the original programs don't really run well (e.g. poepoem). Depending on the C compiler the artefacts remain or disappear.

The most interesting programs from this action:

- freemem (because it shows a nonsensical number; identical over all three; fpc does better but still not quite right)
- poepoem (because it sometimes shows just a white window, no text; differs per compiler; fpc shows the text)
- printcal (because the font size for month varies...; fpc has the same artefact)

Apart from the compilers:

- blokout* (especially blokout2 to show mouse handling when mouse is outside the active window)
- minmax* (hard/hard/fuzzy borders; can depend on video driver?)
- (printcal) (font size for month)

4 Compiling with fpc

See the readme; I used the commandline (not the IDE). At this point there's no Makefile.

5 FPC limitations

A few limitations should be mentioned:

- fpc cannot create OS/2 dll's (shared libraries). In order to create those I used Virtual Pascal 2.1)
- fpc needs the emx dll
- the pwmin unit had/has an error, but my code compensates for that (poormenu, grafmenu, hexcalc2). This error has already been fixed in the trunk.

Some more findings about fpc compiler for OS/2, mostly based on 3.2.2:

- note xxx variables
- missing mousemsg/charmsg/commandmsg (see code in ch00)
- missing conversions from mp to word etc. (see code in ch00)
 - some of this is found but inconsistent
- many missing "macro's" (see my file mousutil.pas in ch00)
- strange insert of menuitem but fixes 3 programs (poormenu, grafmenu, hexcalc2)
- missing constants
- constants with values that conflict with range checks (see 14/head)

6 Program limitations

A short list:

- freemem (because it does not return a correct number but something too large)
- taquin (may not work with smallest cell size; this is specific for the Pascal version)
- printcal (for the varying font size for month: sometimes correct, sometimes way too small)

7 Debug switches

Most if not all of the code should compile and run with range checks enabled (option -Cr).

Range checks must be disabled for these or comment out the offending statements:

- 15/hexcalc2; 06/bitcat1? 06/bitcat2? 06/bricks ?
- 06/bitcat1: pbmi^.argbColor[1].bBlue := \$FF ; etc
- 06/bitcat2: pbmi^.argbColor[1].bBlue := \$FF ; etc
- 06/bricks : pbmi^.argbColor[1].bBlue := \$FF ; etc
- 15/hexcalc2: pacct^.aaccel[1].fs := AF_VIRTUALKEY or AF_SHIFT ; etc

8 Tool set

Compilers:

- fpc 3.2.2
- Virtual Pascal 2.1.279 (for creating DLLs)
- RC 5.00.007 (Jan 30 2003) (needed for the hexcalc programs only)

9 Revisions

20261002 first version.